```
#include <iostream>
#include <string>
using namespace std;

struct Node
{
    string item;
    int quantity, price;
    Node *next;
};

Node *head = NULL;

// Insert Item
void insertItem()
{
    Node *newNode = new Node;

    cout << "Enter item: ";
    cin >> newNode->item;

    cout << "Enter quantity: ";
    cin >> newNode->quantity;

    cout << "Enter price: ";
    cin >> newNode->price;

    newNode->next = NULL;
```

```
    if (head == NULL)
        head = newNode;
    else
    {
        Node *temp = head;

        while (temp->next != NULL)
            temp = temp->next;

        temp->next = newNode;
    }

    cout << "Item added successfully.\n";
}

// Delete Item
void deleteItem()
{
    string item;
    cout << "Enter item to delete: ";
    cin >> item;

    Node *temp = head, *prev = NULL;

    while (temp != NULL && temp->item != item)
    {
        prev = temp;
        temp = temp->next;
    }
```

```
    if (temp == NULL)
    {
        cout << "Item not found.\n";
        return;
    }

    if (prev == NULL)
        head = temp->next;
    else
        prev->next = temp->next;

    delete temp;
    cout << "Item deleted successfully.\n";
}

// Search Item
void searchItem()
{
    string item;
    cout << "Enter item to search: ";
    cin >> item;

    Node *temp = head;

    while (temp != NULL)
    {
        if (temp->item == item)
        {
```

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            cout << "Quantity: " << temp->quantity << endl;
            cout << "Price: Rs. " << temp->price << endl;
            return;
        }

        temp = temp->next;
    }

    cout << "Item not found.\n";
}

// Display Items
void displayItems()
{
    Node *temp = head;

    if (head == NULL)
    {
        cout << "List is empty.\n";
        return;
    }

    while (temp != NULL)
    {
        cout << "\nItem: " << temp->item;
        cout << "\nQuantity: " << temp->quantity;
        cout << "\nPrice: Rs. " << temp->price;
        cout << "\nItem Total: Rs. "
             << temp->quantity * temp->price << endl;
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        if (temp->quantity <= 2)
            cout << "Low Stock Alert!\n";

        temp = temp->next;
    }
}

// Calculate Total Bill
void calculateTotal()
{
    Node *temp = head;
    int total = 0;

    while (temp != NULL)
    {
        total += temp->quantity * temp->price;
        temp = temp->next;
    }

    cout << "Grand Total: Rs. " << total << endl;
}

// Update Quantity
void updateQuantity()
{
    string item;
    cout << "Enter item name: ";
    cin >> item;
```

```
    Node *temp = head;

    while (temp != NULL)
    {
        if (temp->item == item)
        {
            cout << "Enter new quantity: ";
            cin >> temp->quantity;

            cout << "Quantity updated successfully.\n";
            return;
        }

        temp = temp->next;
    }

    cout << "Item not found.\n";
}

// Count Items
void countItems()
{
    Node *temp = head;
    int count = 0;

    while (temp != NULL)
    {
        count++;
```

```
        temp = temp->next;
    }

    cout << "Total types of items: " << count << endl;
}

// Main Function
int main()
{
    int choice;

    do
    {
        cout << "\n--- Stationery Item Management ---\n";
        cout << "1. Insert Item\n";
        cout << "2. Delete Item\n";
        cout << "3. Search Item\n";
        cout << "4. Display Items\n";
        cout << "5. Calculate Total Bill\n";
        cout << "6. Update Quantity\n";
        cout << "7. Count Items\n";
        cout << "8. Exit\n";

        cout << "Enter choice: ";
        cin >> choice;

        switch (choice)
        {
            case 1: insertItem(); break;
```

```
            case 2: deleteItem(); break;
            case 3: searchItem(); break;
            case 4: displayItems(); break;
            case 5: calculateTotal(); break;
            case 6: updateQuantity(); break;
            case 7: countItems(); break;
            case 8: cout << "Program ended.\n"; break;
            default: cout << "Invalid choice.\n";
        }

    } while (choice != 8);

    return 0;
}
```